Cycle Test 1: worked solutions

19 Aug 2025 · 20 marks · 1 hour · CEPC12

Part A (1 mark each)

Q1

1 mark

> What are the different types of fluid flow?

Steady / unsteady; uniform / non-uniform; laminar / turbulent; compressible / incompressible; rotational / irrotational; one-, two- and three-dimensional flow.

Q2

1 mark

> What is capillarity? Give one example.

Capillarity is the rise or fall of a liquid in a narrow tube relative to the surrounding liquid surface, caused by **surface tension** together with adhesion and cohesion: $h = 4\sigma\cos\theta/(\rho g d)$. Examples: **water rising in a thin glass tube** or in soil pores and through a wall (rising damp), oil rising in a lamp wick; mercury shows capillary **depression**.

Q3

1 mark

> State the conditions for stability of submerged surfaces (bodies).

For a **fully submerged** body, the centre of buoyancy B is fixed relative to the body, so stability depends on the relative positions of B and the centre of gravity G:

- **Stable**: B is **above** G
- **Unstable**: B is **below** G
- **Neutral**: B and G **coincide**

Q4

1 mark

> Define the hydrostatic law of pressure distribution.

In a fluid at rest, **pressure increases linearly with depth** at a rate equal to the specific weight: $dp/dz = -\rho g$, i.e. $\mathbf{p = \rho g h}$ below the free surface (gauge). Points at the same level in the same continuous fluid have the same pressure.

Q5

1 mark

> Differentiate between absolute pressure and gauge pressure.

Absolute pressure	Gauge pressure
Measured relative to **absolute zero (perfect vacuum)**	Measured relative to **local atmospheric pressure**
Always positive	Positive, or negative (vacuum)
$p_{abs} = p_{atm} + p_{gauge}$	$p_{gauge} = p_{abs} - p_{atm}$; read by pressure gauges and manometers

Part B (3 marks each)

Q6

3 marks

> A main pipe of diameter 0.4 m carries water at 3 m/s before it branches into two smaller pipes. The first branch (0.25 m) carries water at 4 m/s; the second branch is 0.20 m in diameter. Assuming steady incompressible flow with no leakage, find 1. the flow rate in the main pipe, 2. the discharge in the first branch, 3. the velocity in the second branch.

1. $Q = A\,V = \pi/4 \times 0.4^2 \times 3 =$ **$0.377\ m^3/s$**

2. $Q_1 = \pi/4 \times 0.25^2 \times 4 =$ **$0.196\ m^3/s$**

3. Continuity at the junction: $Q = Q_1 + Q_2 \rightarrow Q_2 = 0.377 - 0.196 = 0.1806\ m^3/s$

$V_2 = Q_2 / (\pi/4 \times 0.2^2) = 0.1806 / 0.03142 =$ **$5.75\ m/s$**

Q7

3 marks

> Two large parallel plane surfaces are 2.4 cm apart and the gap is filled with glycerine. A very thin plate of area 0.5 m^2 is dragged between them at 0.6 m/s. Find the force required if (i) the plate is exactly at the mid-plane, (ii) the plate is 0.8 cm from one surface. $\mu = 8.10 \times 10^{-1}$ N·s/m^2.

The plate is sheared on **both faces**; on each face $F = \mu A V / y$.

(i) Mid-plane ($y = 1.2$ cm on each side):

F on each side = $0.81 \times 0.5 \times 0.6 / 0.012 = 20.25$ N

Total F = $2 \times 20.25 =$ **40.5 N**

(ii) 0.8 cm from one surface (1.6 cm on the other side):

$F_1 = 0.81 \times 0.5 \times 0.6 / 0.008 = 30.38$ N; $F_2 = 0.81 \times 0.5 \times 0.6 / 0.016 = 15.19$ N

Total F = **45.56 N**

The force is least when the plate is at mid-plane.

Q8

3 marks

> A U-tube differential manometer connects pipes A and B. Pipe A contains carbon tetrachloride (S = 1.594) at 11.772 N/cm^2; pipe B contains oil (S = 0.8) at 11.772 N/cm^2. Pipe A lies 2.5 m above pipe B. Find the difference of pressure measured by mercury as the fluid filling the U-tube.

No figure is given; the standard arrangement is assumed: the datum X–X is taken at the **lower mercury surface** (in the limb connected to A), the centre of **pipe B lies on this datum**, and the mercury in B's limb stands h above it.

$p_A = p_B = 11.772$ N/cm^2 = 117,720 N/m^2.

Pressure at X–X, left limb (A side) = $p_A + \rho_1 g\,(2.5) = p_A + 1594 \times 9.81 \times 2.5$

Pressure at X–X, right limb (B side) = $p_B - \rho_2 g\,h + \rho_{Hg} g\,h = p_B + (13{,}600 - 800) \times 9.81 \times h$

Equating ($p_A = p_B$): $1594 \times 2.5 = 12{,}800\,h$

h = 3985 / 12,800 = **0.3113 m = 31.13 cm of mercury**

Although the pipe pressures are equal, the manometer shows a 31.1 cm difference because pipe A is 2.5 m higher and its liquid column is heavier.

Q9

3 marks

> A box-shaped barge of length 50 m, breadth 20 m and draft 3.5 m floats in seawater (1025 kg/m^3). KG of the lightship = 3.0 m. A slack fuel tank 12 m × 6 m, half-filled with diesel (900 kg/m^3), lies on the centreline. A 120 t deck cargo is placed 8.0 m above the keel. Displacement Δ = 5000 t. Find: 1. GM including the free-surface correction; 2. the maximum KG for positive initial stability; 3. the new GM after the cargo is loaded; 4. whether the vessel remains stable.

Note on data: a 50 × 20 × 3.5 m box displaces only 50 × 20 × 3.5 × 1.025 = 3587.5 t, not 5000 t. As intended by the question, the **draft (3.5 m) is used for KB and BM**, and **Δ = 5000 t** for the weight and free-surface calculations.

Hydrostatics of the box (T = 3.5 m):

KB = T/2 = 1.75 m; BM = I/V = (L B^3/12)/(L B T) = B^2/(12T) = 20^2/(12 × 3.5) = 9.524 m

KM = KB + BM = **11.274 m**

Free-surface correction (tank 12 m long × 6 m wide; i = l b^3/12):

i = 12 × 6^3 / 12 = 216 m^4

FSC = ρ_{fuel} × i / Δ = 0.9 × 216 / 5000 = **0.039 m**

1. GM: GM_{solid} = KM − KG = 11.274 − 3.0 = 8.274 m; **GM_{fluid} = 8.274 − 0.039 = 8.235 m**

2. Maximum KG (GM_{fluid} = 0): KG_{max} = KM − FSC = 11.274 − 0.039 = **11.235 m**

3. After loading 120 t at 8.0 m:

New Δ = 5120 t; new KG = (5000 × 3.0 + 120 × 8.0)/5120 = 15,960/5120 = **3.117 m**

Sinkage = 120 / (50 × 20 × 1.025) = 0.117 m → T = 3.617 m

KB = 1.809 m; BM = 400/(12 × 3.617) = 9.215 m; KM = **11.024 m**

FSC = 0.9 × 216 / 5120 = 0.038 m

GM = 11.024 − 3.117 − 0.038 = **7.87 m**

4. GM is **positive (7.87 m)**, so the barge **remains stable**; the cargo raised G by 0.117 m and reduced GM by about 0.37 m. (A box barge this wide has a very large BM, so its GM is large.)

Q10

3 marks

> A rectangular plane surface 1 m wide and 3 m deep (long) lies in water inclined at 30° to the free surface. Its upper edge is 2 m below the surface. Find the total pressure and the centre of pressure.

$A = 1 \times 3 = 3\ m^2$; $\bar{h} = 2 + (3/2)\sin 30° = 2 + 0.75 =$ **2.75 m**

Total pressure $F = \rho g A \bar{h} = 1000 \times 9.81 \times 3 \times 2.75 =$ **80.93 kN**

$I_G = b\,d^3/12 = 1 \times 3^3/12 = 2.25\ m^4$

$h^* = \bar{h} + I_G \sin^2\theta/(A\bar{h}) = 2.75 + 2.25 \times 0.25 / (3 \times 2.75) = 2.75 + 0.068$

Centre of pressure: **h* = 2.818 m below the free surface**